

HAS-MERA-260922 — MERA v1.0 Hardware Architecture Specification

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문서 목적

MERA v1.0은 deterministic multi-channel RF event capture를 위한 architecture baseline이다. AXI4-Stream input, AXI4-Lite control, AXI4-MM output을 기본으로 한다.

Normative Requirements

ID	Priority	Requirement
REQ-001	MUST	AXI4-Stream handshake 준수
REQ-002	MUST	ARMED 상태에서 지정 입력률 흡수
REQ-003	MUST	64-bit trigger timestamp latch
REQ-004	MUST	독립 pre/post length
REQ-005	MUST	channel sample-index alignment 결과 기록
REQ-006	MUST	versioned record header
REQ-007	MUST	overflow/DMA error sticky status
REQ-008	MUST	unique Event ID
REQ-009	MUST	ARM/DISARM/TRIGGER/FLUSH CSR
REQ-010	MUST	invalid configuration 사전 검출
REQ-011	SHOULD	4/8/16 channel
REQ-012	SHOULD	128/256/512-bit AXIS
REQ-013	SHOULD	external/software trigger

Parameter

Parameter	Default	Range
N_CH	8	4/8/16
IQ_BITS	16	8~24
AXIS_W	128	32~512
TS_W	64	fixed
PRE_DEPTH	65536	power-of-two
POST_DEPTH	262144	power-of-two
EVENT_Q_DEPTH	64	16~256
HEADER_BYTES	64	fixed
ADDR_W	64	32/64

Architecture

- AXIS Input Adapter
- Channel Demux / Alignment
- Circular Pre-trigger Buffer
- Trigger Engine
- Capture FSM
- 64-bit Timestamp Counter
- Record Formatter
- Event Descriptor Queue
- AXI4-MM Burst Writer

- CSR / Status / Telemetry

Record Header

Offset	Bytes	Field
0x00	4	MAGIC
0x04	2	VERSION
0x06	2	HEADER_LEN
0x08	4	EVENT_ID
0x0C	4	CHANNEL_MASK
0x10	8	TIMESTAMP
0x18	4	PRE_SAMPLES
0x1C	4	POST_SAMPLES
0x20	4	SAMPLE_WIDTH
0x24	4	SAMPLE_FORMAT
0x28	4	ALIGN_STATUS
0x2C	4	STATUS
0x30	8	PAYLOAD_BYTES
0x38	8	RESERVED

FSM

State	Function
IDLE	capture off
ARMED	circular buffer running
TRIGGERED	timestamp/event id latch
POST_CAPTURE	post samples collect
COMMIT	descriptor creation
DMA_WAIT	DDR burst write
DONE	status/IRQ
ERROR	sticky error

Verification

- Random trigger 100,000 cases, bit-exact golden model.
- AXI protocol assertions.
- Wrap-boundary tests.
- 4/8/16 channel tests.
- $\pm 1/\pm 4$ sample channel offset alignment tests.
- DDR write error injection.
- FPGA timing and sustained bandwidth measurement.

Open Questions

- VR1952 exact RF-ADC→PL physical interface details require product-specific documentation.
- AMD public product page vs DS950 has Channelizer/FFT count discrepancies; DS950 is used for current resource accounting.
- Exact absence of an equivalent AMD confidential/customer IP cannot be proven from public sources.
- DDR sustained bandwidth is board/system dependent.

References

- AMD Versal RF Series Product Page — <https://www.amd.com/en/products/adaptive-socs-and-fpgas/versal/rf-series.html>

- AMD DS950 — Versal Architecture and Product Data Sheet —
<https://docs.amd.com/api/khub/documents/BY0FMkLH05y85Nwu4GmWsw/content>
- AMD DS945 — Versal XQ Architecture and Product Data Sheet —
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- AMD AM009 — AI Engine Architecture Manual: Stream Switch Buffering —
<https://docs.amd.com/r/en-US/am009-versal-ai-engine/Stream-Switch-Buffering-and-Latency>
- AMD AM009 — AI Engine to PL Interface —
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- AMD AM009 — AXI4-Stream Interconnect — <https://docs.amd.com/r/en-US/am009-versal-ai-engine/AXI4-Stream-Interconnect>
- AMD PG022 — AXI DataMover — https://docs.amd.com/r/en-US/pg022_axi_datamover/Features
- AMD/Xilinx RFSoc-MTS GitHub — <https://github.com/Xilinx/RFSoc-MTS>
- AMD/Xilinx RFDC interrupt example —
https://github.com/Xilinx/embeddedsw/blob/master/XilinxProcessorIPLib/drivers/rfdc/examples/xrfdc_intr_example.c